

CHANDRASHEKHAR CHOUDHARY

Cincinnati, Ohio | 513-331-8645 | choudhcr@mail.uc.edu | www.linkedin.com/in/chandrashekharchoudhary/ | [Google Scholar](https://scholar.google.com/citations?user=8Q8Q8Q8Q8Q) | [ORCiD](https://orcid.org/0000-0001-5133-1865)

SUMMARY

Ph.D. candidate in Electrical Engineering specializing in sensor and system design, MEMS, microfabrication, low-power system optimization, and PCB design. 7+ years of experience in microcontroller programming, hardware-software integration, real-time systems design, and cross-functional collaboration. Proven track record of delivering battery-efficient wireless sensor systems, optimizing embedded firmware, and integrating multiple communication protocols.

EDUCATION

Ph.D in Electrical Engineering (Sensor and System Design Track)

University of Cincinnati (UC), Cincinnati, Ohio

April. 2026

GPA: 4.0/4.0

Master of Technology (Electronics System Design)

International Institute of Technology - Bangalore (IIIT Bangalore), Bangalore, India

July. 2018

GPA: 3.7/4.0

RESEARCH EXPERIENCE

Miniature Acoustophoretic Nanoparticle Separator

Present

- Engineered a two-stage microfluidic separator for the precise separation of airborne nanoparticles using acoustophoretic principles.
- Performed design simulations and optimizations using COMSOL and MATLAB to enhance device performance and separation efficiency, device fabrication and implementation in progress.

Smart Helmet System for Traumatic Brain Injury (TBI) Detection

Present

- Led the development of a smart helmet wireless sensor system for monitoring and collection of transient impact data (lasting 15-30 milliseconds) for construction workers, sportsmen etc. to help with traumatic brain injury detection and treatment plans.
- Design and assembly of flexible PCB and development of a system designed to maximize battery life using hardware optimization strategies (>30 hrs. over a coin cell with 85 mAh).
- Developed an energy-efficient software system for impact data collection, activating only during impact events and remaining in sleep mode otherwise to optimize power consumption while continuously monitoring for rare events.

High Efficiency Miniature Multi-Tip Corona Charger for Nanoparticle Detector

2023-25

- Design and implementation of a novel miniature indirect unipolar corona charger with multiple discharge sites for high-efficiency nanoparticle charging and detection.
- Optimized charging system efficiency by minimizing electrostatic losses and increasing ion concentration, ensuring uniform ion distribution inside the charger.
- Validated charger performance through experimental results, achieving a substantial reduction in particle loss (<3.25% at 10.55 nm) without degrading charging efficiency, while ensuring a significantly reduced form factor.

Portable System using Low-Cost Optical Sensors for Real Time Monitoring of Particulate Matter Exposure

2022-24

- Developed a portable, low-cost, fully integrated real-time particle exposure protection system (RT-PEPS) for monitoring respirator fit and particle exposure in emission-intensive industries.
- Improved sensor accuracy by addressing moisture condensation issues in exhaled breath sampling, enhancing the reliability of RT-PEPS.
- Refined the form factor through incremental developments, achieving over 80% size reduction from the initial proof of concept, while maintaining similar system lifetime through targeted hardware and software optimizations.

Battery-Powered Wireless Sensor Network for Non-Invasive Monitoring of Water Usage Events in Premise Plumbing Systems

2021-22

- Developed a battery-powered wireless sensor network for non-invasive monitoring of water usage events in premise plumbing systems using acoustic and calorimetric sensing technologies.
- Utilized low-power wireless communication for real-time data collection, providing insights into water consumption patterns.
- Conducted extensive field testing over 30 days to validate the performance and reliability of the designed sensor system.
- Empowered data-driven decision-making for pipe rightsizing to optimize water conservation, enhance energy efficiency, and reduce the likelihood of Legionella bacteria colony growth in residential and commercial buildings.

Non-Invasive Calorimetric Sensor for Waterflow Event Detection in Premise Plumbing Systems

2021-22

- Designed and developed a novel non-invasive calorimetric sensor to detect waterflow events in premise plumbing systems.
- Optimized the sensor to detect extremely low thermal signals by incorporating thermal isolation techniques and strategically enhancing thermal conductivity at key locations, based on simulation-driven insights.
- Seamless integration of the designed sensor with the battery-powered wireless sensor network system.

Implantable Differential Sensor with Passive Wireless Interrogation for In-Situ Early Detection of Periprosthetic Joint Infection

2021-22

- Designed and fabricated an implantable differential sensor with a passive wireless interrogation system, aimed at the early detection of periprosthetic joint infections.
- Contributed to the development and optimization of the sensor readout circuit, ensuring accurate signal acquisition.

Development of Silicon-Based Piezoresistive Pressure Sensors

Spring '21

- Utilized microfabrication techniques and advanced machinery in a clean room environment to fabricate a pressure sensor on a 2" n-type silicon wafer for pressure measurement through resistance variation.
- Implemented microfabrication processes like oxidation, photolithography (Solitec 3000IR Mask Aligner), etching, ion implantation, metallization, e-beam evaporation (Temescal FC-1800 E-Beam Evaporator), surface analysis (Nanometrics 210XP) to create piezo resistors on a silicon wafer, forming a pressure chamber and measuring sensitivity through resistance and voltage changes using probe station (Micromanipulator 6200 Manual Probe Station).
- Performed device testing by cycling pressure, recording resistance, and voltage values to analyze output voltage changes and validate the sensor's functionality and sensitivity.

WORK EXPERIENCE

Cisco Systems

Bangalore, India

- Led development across Cisco's enterprise switching software stack, focusing on programmability, automation, and internal platform tooling used by **50M+** devices globally.
- Delivered **15+** features across **6** releases on IOS-XE software powering **50M+** switches and routers
- Led cross-functional effort to build an internal automated testing platform → reduced org-wide inefficiencies by **97%**
- Resolved **50+** escalated customer issues through deep diagnostics and stable patch delivery
- Built engineer utilization model adopted org-wide → boosted labor productivity by **25%**

Software Engineer Intern

Jan. 2018 – June 2018

- Developed RF profiling application for client fingerprinting before onboarding onto wireless networks
- Integrated security solution with Cisco routers to enhance network security

Software Engineer Intern

May. 2017 – July 2017

- Created web-based dashboard to analyze and visualize customer bug data across Cisco components
- Enabled data-driven decision making for addressing high-bug components

Ingersoll Rand

Bangalore, India

Hardware Intern

Jan. 2016 – June 2016

- Developed hardware simulator and firmware to emulate sensor signals for testing ThermoKing SR4 controller boards

Schnieder Electric

Bangalore, India

Hardware Intern

Jun. 2015 – July 2015

- Created multi-protocol gateway using Raspberry Pi for building automation applications (Modbus/BACnet).

SKILLS

- **Embedded Systems:** FreeRTOS, GDB, Buildroot, Linux device drivers, Firmware development, RTOS concepts, Baremetal programming, QT, PyQT5
- **Hardware:** ARM Architecture, Silicon Labs, Raspberry Pi, Arduino, Microcontrollers, ADC, DAC, Op-Amps, Boost/buck converters, Multiplexers, Semiconductor devices, System/Board Design, Board bring up, Power budgeting, Hardware automation
- **PCB Design:** Eagle, Flexible PCB design and assembly
- **Software:** SolidWorks, COMSOL Multiphysics, MATLAB, Eagle, LabVIEW, NI Multisim, Origin
- **Technical Skills:** MEMS, Sensor Design & Integration, PCB Design and Assembly, System Design, Process & Product Development, Semiconductor Manufacturing
- **Fabrication Techniques:** Micro-EDM, Photolithography, Oxidation, Metallization, Etching, E-beam evaporation, Cleanroom process optimization, 3D Printing, Microfabrication, Polymer Molding & Casting, Electropolishing
- **Characterization Tools:** Scanning Electron Microscope (SEM), Optical Microscopy, X-ray imaging, Surface Profilometry
- **Communication Protocols:** Zigbee, BLE, I2C, SPI, UART, RS-485, Modbus, BACnet
- **Programming Languages:** C, C++, Python
- **Interpersonal Skills:** Mentorship, Leadership, Teamwork, Communication, Problem-Solving, Adaptability, Project Management

PUBLICATIONS AND PATENTS

Peer-Reviewed Conference Papers:

- **C. Choudhary**, T. Wang, A. Jeyakanthan, J. Romer, S. Shah, M. Islam, E. Nauman, J. Shi, and T. Li, "A smart helmet sensor system for continuous monitoring of transient impact events in low power modes," *IEEE Sensors 2025 Conference*, Vancouver, Canada, Oct. 2025
- **C. Choudhary** and T. Li, "A high efficiency miniature multi-tip corona charger for nanoparticle detectors," *The 38th IEEE Int. Conf. Micro Electromechanical Syst. (MEMS 2025)*, Kaohsiung, Taiwan, Jan. 2025, pp. 1039-1042.
- **C. Choudhary**, G. Batra, S.G. Buchberger, and T. Li, "Non-invasive calorimetric sensor for waterflow event detection in premise plumbing systems," *IEEE Sensors 2022 Conference*, Dallas, TX, Oct. 2022, pp. 1-4.
- **C. Choudhary**, G. Batra, T. Wang, T. Omaghomi, S.G. Buchberger, and T. Li, "Battery-powered wireless sensor network for non-invasive monitoring of water usage events in premise plumbing systems," *IEEE Sensors 2022 Conference*, Dallas, TX, Oct. 2022, pp. 1-4.
- J. Jiang, C. Napier, **C. Choudhary**, H.C. Sagi, C.-Y. Lin, M.T. Archdeacon, and T. Li, "An implantable differential sensor with passive wireless interrogation for in-situ early detection of periprosthetic joint infection," *The 36th IEEE Int. Conf. Micro Electro Mechanical Syst. (MEMS 2023)*, Munich, Germany, Jan. 2023, pp. 235-238.
- J. Jiang, K. Sureshkumar, **C. Choudhary**, T. Kerkes, H.C. Sagi, C.-Y. Lin, M.T. Archdeacon, and T. Li, "A passive wireless differential sensor for in-situ early detection of periprosthetic joint infection," *Solid State Sensors, Actuators, Microsyst. Workshop (Hilton Head 2022)*, Hilton Head Island, SC, Jun. 2022, pp. 5-8.

Short Conference Abstracts and Presentations:

- J. Wang, T. Li, **C. Choudhary**, X. Niu, M. Yermakov, and S. Yermakov, "Optical-based Measurement of Workplace Protection Factor for Aerosols Filtering Respirators," *The European Aerosol Conference 2024*, Tampere, Finland, Aug. 2024.
- T. Li, J. Jiang, K. Sureshkumar, **C. Choudhary**, T. Kerkes, H.C. Sagi, C.-Y. Lin, and M.T. Archdeacon, "An implantable passive wireless differential sensor for in-situ early detection of periprosthetic joint infection," *IEEE Sensors Conference – Open Poster*, Oct. 2023
- J. Jiang, T. Li, K. Sureshkumar, **C. Choudhary**, T. Kerkes, H.C. Sagi, C.-Y. Lin, and M.T. Archdeacon, "An Implantable Differential Sensor with Passive Wireless Interrogation for In-Situ Early Detection of Periprosthetic Joint Infection," *IRIS Ignite Annual Expo & Festival of Sensing*, University of Cincinnati, OH, May 2022

Patents Pending and Invention Disclosures:

- **C. Choudhary** and T. Li, *Method for Continuous Monitoring of Rare Transient Events*, U.S. Provisional Patent Application No. 63/896,932, submitted Oct. 2024.
- T. Li, C.H. Ahn, M.S. Islam, **C. Choudhary**, S. Setty, T. Wang, L. Varsha, J. Shi, and E. Nauman, *Smart Monitoring Device and System for Safety Helmets*, U.S. Provisional Patent Application No. 63/864,858, submitted Aug. 2025.
- T. Li and **C. Choudhary**, *Device for Charging and Detecting Nanoscale Particulate Matters*, U.S. Provisional Patent Application No. 63/737,144, submitted Dec. 2024.

- J. Wang, T. Li, M. Yermakov, S. Grinshpun, **C. Choudhary**, X. Niu, T. Wang, and S.R. Dafe, *A Device for Better Respirator Usage and Monitoring*, U.S. Provisional Patent Application No. 63/688,263, submitted Aug. 2024.
- T. Li, S.G. Buchberger, **C. Choudhary**, G. Batra, T. Omaghomi, *Water Usage Detection and Logging in Plumbing Systems*, U.S. Patent Pending, Application No. 18/495,073, submitted Oct. 2023.
- T. Li, J. Jiang, K. Sureshkumar, **C. Choudhary**, H.C. Sagi, C.-Y. Lin, M. Archdeacon, *Device and Method for Detection of Post-Surgical Infection and Other Disease*, U.S. Patent Pending, Application No. 18/201,179, submitted May 2023.